Aaradhya Dev Tamrakar

AI/ML Engineering · Applied ML · Embedded AI · AI Workflows

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Final-year Electronics & Information Engineering student at IOE Kathmandu Engineering College and Fuse AI Fellow. Builds applied ML systems, embedded AI (ESP32-S3 / TFLite Micro), and AI workflow tools. Uses AI to accelerate development and turns recurring workflows into reusable tools and deterministic checks.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Co-developed **BiasAperture**, an automated demographic fairness auditing pipeline measuring predictive disparities on computer vision benchmarks as the fellowship capstone project.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026

Python · AIF360 · Fairlearn · Statistical Testing · Jinja2 · OpenCV

- Co-developed an automated prediction-auditing and reporting pipeline as a two-person fellowship capstone with Tisha Manandhar.
- Audited classification performance across **126 demographic bins** on the FairFace benchmark (**97,698 images**), computing Disparate Impact Ratio, Demographic Parity Difference, and Equalized Odds.

SPARK — Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TensorFlow Lite Micro · Python · FreeRTOS · MPU-6050

- Project lead for ML modeling, gateway processing, and explainability in a four-person team; hardware and firmware have dedicated owners.
- Trained 1D-CNN fall-detection models on the SisFall benchmark (38 subjects, 38,420 windows), evaluating subject-stratified cross-validation with **0.9185 AUC-ROC** and 87.81% sensitivity.

Fleet Orchestrator — AI Workflow Coordination & Tooling | 2025 – Present

Python · FastAPI · Provider Adapters · Model Context Protocol (MCP) · SQLite

- Built coordination tooling for multi-step AI workflows, with task state, worker adapters, scheduling, and live status inspection.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Coordinate branch planning, technical workshops, and cross-committee communication for Kathmandu Engineering College IEEE student body.
- Facilitated technical awareness events and student engagement initiatives across engineering departments.

TECHNICAL SKILLS

Machine Learning & Evaluation: Python, PyTorch, scikit-learn, Pandas, NumPy, XGBoost, SHAP, Fairlearn, AIF360, OpenCV, MediaPipe

AI Systems, Backend & Tooling: FastAPI, Docker, Model Context Protocol (MCP), SQL (PostgreSQL, SQLite), asyncio, REST APIs, MLflow

Embedded Systems & Edge AI: C/C++, ESP32-S3, TensorFlow Lite Micro, FreeRTOS, Arduino, Sensor Interfacing (IMU, UART/I2C/SPI), Kotlin

Engineering Practice & Tooling: Git, GitHub, Linux, PowerShell, Pytest, Reproducible Configurations, WordCraft, LaTeX